

1. Project Overview

Business Problem

As a customer-operations analyst, the objective is to provide Support Leadership with actionable insights to distinguish between persistent, long-term issues and rapidly emerging problems within the support ecosystem.

Scope & Tooling

Dataset: Customer Support Ticket Dataset (Kaggle).

Tools: Excel/Power Query, Python/Pandas, SQL, Power BI.

Note: Ticket category serves as an observed issue classification; identifying true root causes may require merging with product/process telemetry.

2. Data Preparation & Cleaning Workflow

To ensure reporting accuracy, the raw Kaggle dataset was processed using the following systematic "baby-step" data cleaning procedures prior to SQL ingestion:

1. **Validate Ticket ID Uniqueness:** Checked for duplicated primary keys to prevent double-counting.
2. **Date Standardization:** Converted all raw ticket submission and closure dates to standard datetime formats.
3. **Categorical Standardization:** Normalized casing and spelling for ticket type, status, priority, and channel fields.
4. **Null Value Inspection:** Investigated missing resolution and satisfaction values (e.g., leaving them NULL for pending tickets rather than forcing a 0).
5. **Duration Calculation:** Converted time-to-resolution metrics into standard numeric duration fields (hours/days).
6. **Logical Validation:** Checked for negative or impossible durations (e.g., resolution times preceding creation times).
7. **Temporal Grouping:** Created derived month and week fields to enable time-series trending.
8. **SLA Flags:** Created binary SLA breach flags where a documented Service Level Agreement exists.
9. **Metric Separation:** Kept open/pending ticket metrics explicitly separate from resolved-ticket historical metrics to avoid skewing average resolution times.

Database Setup: Initialized the environment using:

```
CREATE DATABASE support_ticket_rootcauseanalysis;  
USE support_ticket_rootcauseanalysis;
```

3. Diagnostic SQL Queries & Business Insights

Q1. Which products generate the highest number of customer support tickets?

```
SELECT product_purchased, COUNT(ticket_id) AS total_tickets FROM support_ticket_RCA  
GROUP BY product_purchased ORDER BY COUNT(ticket_id) DESC;
```

Business Insight: Identifying the products with the highest ticket volume highlights where engineering improvements or enhanced self-service resources are most urgently needed.

Q2. What are the most common types of customer issues?

```
SELECT ticket_type, COUNT(ticket_id) AS total_tickets FROM support_ticket_RCA GROUP  
BY ticket_type ORDER BY COUNT(ticket_id) DESC;
```

Business Insight: Categorizing ticket volume by issue type allows support operations to allocate specialized agents to the most frequent customer friction points.

Q3. How many tickets are Open, Pending, and Resolved?

```
SELECT ticket_status, COUNT(ticket_id) AS total_tickets FROM support_ticket_RCA  
GROUP BY ticket_status ORDER BY COUNT(ticket_id);
```

Business Insight: Monitoring the volume of Open and Pending versus Resolved tickets provides a rapid health check on the current support backlog and agent bandwidth.

Q4. How many tickets exist for each priority level?

```
SELECT ticket_priority, COUNT(ticket_id) AS total_tickets FROM support_ticket_RCA  
GROUP BY ticket_priority ORDER BY COUNT(ticket_id) DESC;
```

Business Insight: Tracking ticket priority distribution ensures that critical, system-breaking issues are properly segmented and not buried beneath standard inquiries.

Q5. Which support channel receives the most customer tickets?

```
SELECT ticket_channel, COUNT(ticket_id) AS total_tickets FROM support_ticket_RCA
GROUP BY ticket_channel ORDER BY COUNT(ticket_id) DESC;
```

Business Insight: Analyzing ticket origin channels informs workforce management, ensuring adequate staffing for high-traffic, real-time mediums like live chat or phone support.

Q6. What is the average resolution time for each ticket priority/resolution type?

```
SELECT TOP 10 resolution, ROUND(AVG(resolution_hours), 2) AS avg_resolution_hours
FROM support_ticket_RCA WHERE resolution_hours IS NOT NULL GROUP BY resolution
ORDER BY ROUND(AVG(resolution_hours), 2) DESC;
```

Business Insight: Evaluating average resolution times against specific categorizations identifies complex process bottlenecks and helps leadership set realistic SLA benchmarks.

Q7. What is the average customer satisfaction rating for each ticket type?

```
SELECT ticket_type, ROUND(AVG(customer_satisfaction_rating), 2) AS
avg_satisfaction_rating FROM support_ticket_RCA WHERE customer_satisfaction_rating
IS NOT NULL GROUP BY ticket_type ORDER BY ROUND(AVG(customer_satisfaction_rating),
2) DESC;
```

Business Insight: Measuring satisfaction across ticket types reveals which specific technical or billing issues are causing the most lasting frustration for our customer base.

Q8. Which support channels have the highest and lowest average customer satisfaction ratings?

```
WITH channel_ratings AS ( SELECT ticket_channel,
ROUND(AVG(customer_satisfaction_rating), 2) AS avg_satisfaction_rating FROM
support_ticket_RCA WHERE customer_satisfaction_rating IS NOT NULL GROUP BY
ticket_channel ) SELECT * FROM channel_ratings WHERE avg_satisfaction_rating =
(SELECT MAX(avg_satisfaction_rating) FROM channel_ratings) OR
avg_satisfaction_rating = (SELECT MIN(avg_satisfaction_rating) FROM
channel_ratings);
```

Q9. Which products have the largest number of unresolved tickets?

```
SELECT product_purchased, COUNT(ticket_id) AS unresolved_tickets FROM
support_ticket_RCA WHERE ticket_status IN ('Open', 'Pending') GROUP BY
product_purchased ORDER BY unresolved_tickets DESC;
```

Business Insight: Pinpointing products with the highest unresolved ticket counts alerts product managers to potential systemic defects or complex bugs requiring engineering escalation.

Q10. Which product and ticket-type combinations have the highest number of tickets and lowest customer satisfaction?

```
SELECT product_purchased, ticket_type, COUNT(ticket_id) AS total_tickets,
ROUND(AVG(customer_satisfaction_rating), 2) AS avg_satisfaction_rating FROM
support_ticket_RCA GROUP BY product_purchased, ticket_type ORDER BY total_tickets
DESC;
```

Business Insight: Cross-referencing high-volume product/issue combinations with poor satisfaction scores highlights the most critical "danger zones" requiring immediate cross-functional intervention.

4. Recommended Power BI Dashboard Output

To operationalize these SQL findings for leadership, the following dashboard architecture is recommended:

Page 1: Executive KPI Overview

- **Metric Cards:** Total Tickets, Open Backlog, Average Resolution Time (SLA Breaches), Average CSAT.
- **Trend Line Chart:** Ticket volume month-over-month (Addresses Q1 & Q3).
- **Donut Charts:** Volume distribution by Channel (Q5) and Priority level (Q4).

Page 2: Diagnostic Deep-Dive (Drill-Down)

- **Bar Chart Matrix:** Ticket volume by Category vs. Average Resolution Time (Addresses Q2 & Q6).
- **Scatter Plot:** Issue Volume vs. Average CSAT score by Product category (Addresses Q10).
- **Decomposition Tree:** Root cause breakdown branching from Product → Channel → Ticket Type → Status (Open/Resolved) (Addresses Q9).